

A Decoder-Only Jacobian Lens for Speech-to-Text Models

Januda Lelwala, Anas Hussaindeen, Chandupa Ambepitiya, and Dewmike Amarasinghe
Department of Computer Science and Engineering
University of Moratuwa, Sri Lanka

Abstract—Encoder–decoder speech recognizers such as Whisper fuse acoustic evidence into a decoder residual stream that is then unembedded as tokens. We ask whether the Jacobian lens—introduced to identify verbalizable representations in language models—transfers to this decoder. A natural first attempt, a pooled encoder-to-decoder transport, destroys the causal (t, t') structure of the map and yields scores without logit semantics. We instead fit one averaged, position-resolved Jacobian per decoder layer, $J_\ell = \mathbb{E}_{t, t' \geq t} [\partial h_{\text{final}, t'} / \partial h_{\ell, t}]$, estimated with Hutchinson probes, and read out with the model’s own output projection: $\text{lens}(h) = \text{softmax}(E J_\ell h)$. The resulting (position $\times$ layer) grid is implemented in ECHO, an audio interpretability workbench. Qualitative readings recover crystallization depth, an audio-only workspace at the start-of-transcript token, language-prior versus acoustic evidence, and precursor signatures of hallucination loops.

Index Terms—speech recognition, mechanistic interpretability, Jacobian lens, Whisper, encoder–decoder Transformers

I. INTRODUCTION

Automatic speech recognition (ASR) models such as Whisper [1] are now accurate enough that *how* they transcribe matters as much as *what* they transcribe. Hallucinated repetitions under silence, homophone substitutions, and early cutoffs are familiar failure modes, yet the internal state that produces the next token is rarely inspected in vocabulary space.

A parallel question has just been posed for language models. Gurnee et al. [2] introduce the *Jacobian lens*: an average causal map from a layer’s residual stream to the final pre-logit state, read out with the model’s own unembedding. The tokens that this readout ranks are the representations the model is *poised to verbalize*. In LLMs those representations occupy a mid-depth band with the functional hallmarks of a global workspace.

Speech-to-text models look, at first glance, like a hard case for the same construction. The residual stream is split across two components. Acoustic computation lives in an encoder that is never unembedded; verbalization lives in a decoder that attends to the encoder through cross-attention [3]. The question is which stream the lens should read.

This short paper answers: the decoder, and only the decoder. We port the LLM Jacobian-lens estimator to Whisper’s decoder residual stream, fit it with Hutchinson probes [4] so that the encoder can sit outside the autograd graph, and ship the result as a first-class tool in ECHO, a Learning Interpretability Tool for audio models [5]. Along the way we record why the obvious alternative—a pooled transport from encoder blocks into the decoder—is the wrong object.

Contributions:

- A decoder-only J-lens for seq2seq ASR that matches the LLM construction: same-space maps, causal averaging over $t' \geq t$, readout $\text{softmax}(E J h)$.
- A failure analysis of pooled encoder transports (destroyed alignment, off-manifold unembedding, scores without probability semantics).
- An implementation in ECHO, with a (position $\times$ layer) visualization and a linear closed-form test, plus qualitative readings of Whisper-base.

II. RELATED WORK

Lenses on language models. The logit lens [6] applies the unembedding to intermediate residuals (identity transport). The tuned lens [7] learns an affine map to the final layer. The Jacobian lens [2] is neither: it is the *average derivative* of the true residual-to-final map, not a regression onto final states. We copy that estimator, not the LLM causal suite (write, hold, ablate), which we do not claim to have reproduced.

Encoder–decoder interpretation. DecoderLens [8] skips encoder layers into a frozen decoder to ask what the encoder has computed. Complementary: they interpret the encoder *through* the decoder; we interpret the decoder *with* its own Jacobian. Linear probes [9]–[11] diagnose linearly available features but are not causal maps onto the model’s unembedding.

Speech tooling. ECHO is the audio analogue of LIT [5]: attention, saliency, embeddings, and perturbations on Whisper and wav2vec 2.0 [12]. Attention says where the decoder looked in the audio; saliency says which waveform regions moved a scalar. Neither says what the decoder is poised to say. CTC models have no decoder residual to unembed and are excluded.

III. METHOD

A. Substrate

Whisper is an encoder–decoder Transformer. For a clip, the encoder emits a length-1500 acoustic sequence (30 s of log-Mel, stride 2). The decoder, at each transcript position t , attends causally to the prefix and, via cross-attention, to all 1500 encoder frames, then unembeds with a tied projection E . On Whisper-base, both stacks have width $d = 512$ and six blocks; the Hugging Face decoder reports seven residual streams (embedding output plus six blocks). The fit target is the model’s last hidden state, the post-final-LayerNorm tensor

that E consumes. We write $h_{\ell,t} \in \mathbb{R}^d$ for the residual at lens-layer ℓ and position t , and $h_{\text{final},t'}$ for that target.

The encoder is a substrate, not a workspace. It never sees E . Gradients for the lens therefore never enter it: one frozen forward provides cross-attention keys and values.

B. Estimator

Following [2], the lens at layer ℓ is the expected causal Jacobian

$$J_\ell = \mathbb{E}_{t, t' \geq t, \text{clip}} \left[\frac{\partial h_{\text{final},t'}}{\partial h_{\ell,t}} \right] \in \mathbb{R}^{d \times d}. \quad (1)$$

The expectation is over source positions, future positions allowed by the causal mask, and a corpus of teacher-forced transcripts. There is no optimizer and no learned baseline: the model is frozen in evaluation mode, and J_ℓ is *measured*.

Exact Jacobians are unnecessary. A Rademacher probe $r \in \{\pm 1\}^d$ and one vector-Jacobian product of $\sum_{t'} \langle h_{\text{final},t'}, r \rangle$ yield, at every source t simultaneously, $\sum_{t' \geq t} J_\ell(t \rightarrow t')^\top r$, because the causal mask zeroes $t > t'$ [4]. Averaging $\text{outer}(r, \sum_t g_t)$ over probes and clips, and dividing by the triangular count $T(T+1)/2$, estimates (1). The encoder sits outside this graph.

C. Readout

At apply time the decoder actually runs: greedy generation, or a provided reference transcript, produces the positions; a teacher-forced pass collects $h_{\ell,t}$. Each cell is

$$\text{lens}(h_{\ell,t}) = \text{softmax}(E J_\ell h_{\ell,t}). \quad (2)$$

Whisper’s final LayerNorm is inside the decoder, so it is absorbed into J_ℓ ; we do not apply an extra RMS-norm. Scores sit on the model’s logit scale. Top- k (default $k = 5$) is a *ranking* of verbalizable content, not a phrase, and the softmax is for display, not calibrated confidence.

The readout axis is transcript position, not audio time. Which encoder frame supports a token remains a cross-attention question.

D. Why encoder transports fail

An earlier design fitted a global map from the mean-pooled encoder state to the mean-pooled decoder state and applied it to 0.3 s encoder buckets (Table I). That object is not (1). Differentiating *after* pooling erases (t, t') alignment; the linearization is around whole-clip means but is applied locally; the unembedding is used off-manifold with the constant term dropped, so scores have no probability semantics. CTC support came for free (the encoder is the only stream) and disappeared with the redesign, correctly: a CTC head never verbalizes through a decoder workspace.

IV. IMPLEMENTATION

The lens is a job in ECHO (Fig. 1). Fit writes a format-v2 artifact: architecture “decoder”, method Hutchinson-decoder-VJP, one square $d \times d$ matrix per recorded decoder state, bound to a model revision. Apply refuses a mismatch. Only seq2seq

TABLE I
ENCODER TRANSPORT VERSUS DECODER-ONLY LENS.

	Pooled encoder transport	Decoder-only J-lens
Source	mean of 1500 encoder frames	residual $h_{\ell,t}$
Target	mean of T decoder positions	$h_{\text{final},t'}, t' \geq t$
Spaces	encoder $\rightarrow$ decoder	decoder $\rightarrow$ decoder
Readout	$(h - \bar{h})J^\top E^\top$	$\text{softmax}(E J h)$
Units	0.3 s buckets	decoder tokens

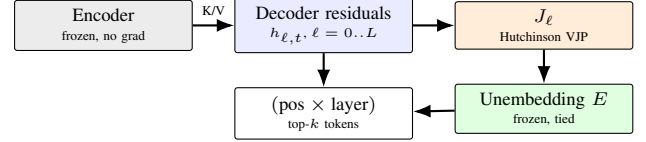


Fig. 1. ECHO decoder-only J-lens. The encoder supplies cross-attention keys and values and is never differentiated. Each decoder layer has its own J_ℓ ; readout uses the model’s E .

ASR adapters expose the architecture; wav2vec 2.0 returns none.

On Whisper-base the grid has seven rows (embedding output plus six blocks). Unit tests recover the closed form $J = 2M/(T+1)$ for a linear decoder with causal triangular averaging, and check apply against an independent $(Jh)E^\top$ recomputation.

V. WHAT THE GRID SHOWS

The following readings are observational, from Whisper-base in the ECHO lab. They are hypotheses to measure, not census results. Layer 0 is the embedding output; layer 6 is the last decoder block.

Crystallization depth. A column (one position, all layers) typically moves from frequent-token rankings in the early rows, through syntactic candidates, to the spoken word in the upper-middle band. The row of convergence is a candidate for the depth at which acoustics enter the verbal decision. Columns that never converge mark transcription against the model’s internal preference.

Start-of-transcript as audio-only workspace. At the start-of-transcript token the decoder has consumed no transcript tokens. Everything in the residual arrived through cross-attention. The top tokens at this position are an acoustic gist of the clip—likely first words, Whisper’s silence convention, or background-speech artifacts—the closest available reading of “what the audio alone put into the workspace.”

Prior versus evidence. When mid and late layers agree on a collocation, the token is language-model driven. When mid layers stay generic and the top rows flip to an unusual word, the token came from the audio. Sweeping that disagreement is a per-word prior-versus-evidence map.

Hallucination precursors. Under silence, Whisper’s repetition loop is often ranked highly in mid layers *before* generation commits. The grid is a leading indicator, not only a post-hoc explanation.

Ambiguity and endpoints. A wide top- k spread at a position (their/there/they’re) flags under-determined acoustics.

Near the end of speech, readouts drift toward end-of-text, punctuation, or trained closers such as “thank you”; premature stop-direction while speech continues is an early-cutoff signature.

These patterns are the speech analogue of the LLM observation that early-layer readouts are noise and that a mid-depth band carries verbalizable content [2]. Confirming a workspace *range* with a layer-wise top-1 agreement curve is left as future work.

VI. LIMITATIONS

The map is a first-order average: rankings are more trustworthy than absolute scores, and a position far from the fit distribution will be misread. Because J_ℓ averages over pairs, the lens cannot name the supporting audio frame. Fit targets are teacher-forced, so J reflects “given this prefix,” not free-running generation. Hutchinson probes (we default to four) leave residual estimation noise. Frequent-token geometry biases rankings. The artifact is model-revision bound. Write-mode interventions (steer, ablate, patch) from [2] are not implemented. We report no large-scale census.

VII. CONCLUSION

The Jacobian lens is a statement about a residual stream that is about to be unembedded. In encoder–decoder ASR that stream is the decoder’s. Fitted there, with causal averaging and the model’s own E , it yields a (position $\times$ layer) vocabulary grid that ECHO can already show. Fitted on pooled encoder states, it yields an object that is no longer a lens. The next measurements—workspace-band statistics, multi-size Whisper, and a write mode—should be taken on the decoder construction, not the encoder one.

ACKNOWLEDGMENTS

We thank Dr. Uthayasanker Thayasivam for guidance, and the ECHO-Lit group for the platform this lens ships in. The estimator follows Gurnee et al. [2]; remaining errors are ours.

REFERENCES

- [1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 28 492–28 518.
- [2] W. Gurnee, N. Sofroniew, A. Pearce, M. Piotrowski, I. Kauvar, R. Chen, A. Soligo, P. Bogdan, E. Ong, R. Wang, B. Thompson, D. Abrahams, S. Kantamneni, E. Ameisen, J. Batson, and J. Lindsey, “Verbalizable representations form a global workspace in language models,” *arXiv preprint arXiv:2607.15495*, 2026.
- [3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [4] M. F. Hutchinson, “A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines,” *Communications in Statistics — Simulation and Computation*, vol. 18, no. 3, pp. 1059–1076, 1989.
- [5] I. Tenney, J. Wexler, J. Bastings, T. Bolukbasi, A. Coenen, S. Gehrmann, E. Jiang, M. Pushkarna, G. Ramaswamy, E. Reif, and A. Yuan, “The language interpretability tool: Extensible, interactive visualizations and analysis for NLP models,” in *Proceedings of EMNLP: System Demonstrations*, 2020, pp. 107–118.
- [6] nostalgebraist, “interpreting GPT: the logit lens,” <https://www.lesswrong.com/posts/AcKRB8wDpdN6v6ru/interpreting-gpt-the-logit-lens>, 2020.
- [7] N. Belrose, Z. Furman, L. Smith, D. Halawi, I. Ostrovsky, L. McKinney, S. Biderman, and J. Steinhardt, “Eliciting latent predictions from transformers with the tuned lens,” in *arXiv preprint arXiv:2303.08112*, 2023.
- [8] A. Langedijk, H. Mohebbi, G. Sarti, W. Zuidema, and J. Jumelet, “DecoderLens: Layerwise interpretation of encoder-decoder transformers,” in *Findings of the Association for Computational Linguistics: NAACL 2024*. Mexico City, Mexico: Association for Computational Linguistics, 2024, pp. 806–819.
- [9] G. Alain and Y. Bengio, “Understanding intermediate layers using linear classifier probes,” in *ICLR Workshop*, 2017.
- [10] J. Hewitt and C. D. Manning, “A structural probe for finding syntax in word representations,” in *Proceedings of NAACL-HLT*, 2019, pp. 4129–4138.
- [11] Y. Belinkov, “Probing classifiers: Promises, shortcomings, and advances,” *Computational Linguistics*, vol. 48, no. 1, pp. 207–219, 2022.
- [12] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 12 449–12 460.